Chemistry GCE A Level 2024 - Paper 1

Duration: 1h30 | Cameroon GCE Board

GENERAL CERTIFICATE OF EDUCATION BOARD, General Certificate of Education Examination. 0715 CHEMISTRY 1, ADVANCED LEVEL, JUNE 2024. Multiple choice question paper. Duration: one and a half hours (1h 30 minutes). Answer ALL the 50 questions in this examination. All questions carry equal marks. Non-programmable calculators are allowed. Each question has FOUR suggested answers: A, B, C and D. Decide which answer is correct and mark only one answer for each question (if more than one answer is marked, the question scores zero). Avoid spending too much time on any one question; if a question is difficult, move on and come back to it later. Mobile phones are NOT allowed in the examination room.

Questions 1 - 35 (Thirty-five questions)

Directions: Each of the questions or incomplete statements in this section is followed by four suggested answers. Select the best answer in each case.

Questions 36 - 45 (Ten questions)

Directions: For each of the questions below, ONE or MORE of the responses is (are) correct. Decide which of the responses is (are) correct. Then choose:

A	B	C	D
1, 2 and 3 correct	1 and 2 only correct	2 and 3 only correct	3 only is correct

- **A** if 1, 2 and 3 are all correct
- **B** if 1 and 2 only are correct
- **C** if 2 and 3 only are correct
- **D** if 3 only is correct

Questions 46 - 50 (Five questions)

Directions: Each of the following questions consists of a statement in the left-hand column followed by a second statement in the right-hand column. Decide whether the first statement is true or false. Decide whether the second statement is true or false. Then choose:

- **A** if both statements are true and the second statement is a CORRECT explanation of the first statement.
- **B** if both statements are true and the second statement is NOT a CORRECT explanation of the first statement.
- **C** if the first statement is true, but the second statement is false.
- **D** if the first statement is false, but the second statement is true.

	First statement	Second statement	
A	True	True	Second statement is a CORRECT explanation of the first

	First statement	Second statement	
B	True	True	Second statement is NOT a CORRECT explanation of the first
C	True	False	
D	False	True	

1. Molar mass is the mass of

a) one molecule of a substance
b) one mole of a substance
c) one atom of a substance
d) $1/12^{\text{th}}$ the mass of one molecule of a substance

2. The mass number of an atom is the sum of the number of

a) protons and neutrons in the nucleus of an atom
b) electrons and protons in an atom
c) electrons, protons and neutrons in an atom
d) protons in the nucleus of an atom.

3. Why are some ionic compounds like AgCl insoluble in water which is a polar solvent?

a) Their lattice energy is higher than their solvation energy
b) Their lattice energy is lower such that it cannot compensate for the solvation energy
c) Their cations have a high charge density which needs high energy to be broken
d) The hydration energy is higher than their lattice energy.

4. The mass of 1 mole of calcium is 40 g. What is the mass of one calcium atom?

a) 6.02×10^{23}
b) 6.02×10^{-23}
c) 6.65×10^{23}
d) 6.65×10^{-23}

5. Nitration of benzene takes place in the presence of a nitrating mixture. Identify the nitrating mixture amongst the following:

a) dilute $\mathrm{H_2SO_4}$ / dilute $\mathrm{HNO_3}$
b) Conc. $\mathrm{H_2SO_4}$ / Conc. $\mathrm{HNO_3}$
c) $\mathrm{NO_2^+}$
d) $\mathrm{HNO_3}$

6. When compound X is heated with concentrated ethanoic acid and concentrated sulphuric acid, compound Y of molecular formula $\mathrm{C_3H_6O_2}$ is formed. The molecular formula of compound X is

a) $\mathrm{CH_3CH_2OH}$
b) $\mathrm{CH_3OH}$
c) $\mathrm{CH_3CH_2CH_2OH}$
d) $\mathrm{(CH_3)_2CO}$

7. $100\ \mathrm{cm^3}$ of $0.10\ \mathrm{M}$ solution of a certain metal chloride required $200\ \mathrm{cm^3}$ of $0.20\ \mathrm{M}$ solution of silver nitrate for complete precipitation of the chlorides. The right formula for the metal chloride could be?

a) MCl_4
b) MCl
c) MCl_3
d) MCl_2

8. H_2O_2 decomposes according to the equation: $2H_2O_2 \rightarrow 2H_2O + O_2$. What is the rate of disappearance of H_2O_2 with respect to the rate of formation of O_2 gas?

a) 2
b) 1
c) 1/2
d) 2/3

9. Identify the energy change(s) involved in the reaction $Li(g) + F(g) \rightarrow Li^+(g) + F^-(g)$

a) Atomization and ionization of Lithium.
b) ionization of Lithium and electron affinity of Fluorine
c) Ionization of Lithium and ionization of Fluorine
d) Lattice energy of Lithium and Fluorine

10. The Bragg's equation is given by $2d \sin\theta = n\lambda$. d in the equation represents:

a) Distance between two atoms
b) Distance between two layers
c) Distance between x-rays
d) Wave length

11. Which of the following oxides is solid and acidic?

a) PbO_2
b) CO_2
c) SnO_2
d) SiO_2

12. Identify the pair of s-block elements that show diagonal relationship.

a) Li/Be
b) B/Si
c) Li/Mg
d) Be/Al

13. What are isomers?

a) Atoms of same element with different atomic mass but same atomic number
b) They are different forms of same element in same physical state
c) Compounds with same molecular formula but different structural formulae
d) Compounds of same elements but different physical states.

14. Catenation as applied to carbon Chemistry is the ability of carbon to

a) form compounds involving only carbon atoms
b) undergo hybridization

c) form strong bonds with oxygen
d) form strong bonds with itself

15. The ground state electronic configuration of nitrogen with atomic number 7 using s p d f notation is:

a) $1s^2\,2s^2\,3p^3$
b) $1s^2\,2s^1\,2p^4$
c) $1s^2\,2s^2\,2p^3$
d) $1s^2\,2s^1\,3p^4$

16. A cell is constructed using the following two half cells:

$$\mathrm{Ni^{2+}(aq) + 2e^- \rightarrow Ni(s)\,;\, E^{\ominus} = -0.25\ V}$$

$$\mathrm{Ag^{+}(aq) + e^- \rightarrow Ag(s)\,;\, E^{\ominus} = +0.80\ V.}$$

Choose the correct cell diagram of the cell.

a) $\mathrm{Ni(s)\,/\,Ni^{2+}(aq)\,//\,Ag^{+}(aq)\,/\,Ag(s)}$
b) $\mathrm{Ag(s)\,/\,Ag^{+}(aq)\,//\,Ni(s)\,/\,Ni^{2+}(aq)}$
c) $\mathrm{Ag(s)\,/\,Ag^{+}(aq)\,//\,Ni^{2+}(aq)\,/\,Ni(s)}$
d) $\mathrm{Ni(s)\,/\,Ni^{2+}(aq)\,//\,Ag(s)\,/\,Ag^{+}(aq)}$

17. The overall reaction

$$\mathrm{(CH_3)_2CHBr + KOH \rightarrow (CH_3)_2CHOH + KBr}$$

is best described as

a) Electrophilic substitution
b) Free radical substitution
c) Nucleophilic Substitution
d) Elimination

18. The sulphates and carbonates of Mg, Ca, and Sr are insoluble or sparingly soluble in water because

a) the metals are hard
b) their lattice energies are higher than the hydration energies of their ions
c) the hydration energies of the ions are larger than their lattice energies
d) the cations have a high charge density

19. Raoult's law is stated as:

a) The vapour pressure above a mixture of liquids is the sum of its various components
b) The partial vapour pressure of a component in a liquid mixture is the product of its mole fraction and the total pressure of the mixture
c) The partial vapour pressure of a component in a liquid mixture is inversely proportional to its mole fraction
d) The partial vapour pressure of a component in a liquid mixture is the product of its mole fraction and its pure vapour pressure

20. Why is it that lithium is the only metal in group I of the periodic table without a 4th ionization energy?

a) It has the smallest atomic radius
b) It is the most unreactive metal in group I

c) It has only three electrons in its atoms
d) It has the highest effective nuclear charge

21. The correct systematic name for the compound $[Co(en)_2Cl_2]NO_3$ is

a) Dichlorobisethylenediamine cobaltate(III) nitrate
b) Bisethylenediaminedichlorocobaltate(III) nitrate
c) Bisethylenediaminedichlorocobalt(III) nitrate
d) Dichlorobisethylenediaminecobalt(III) nitrate

22. Hoffmann degradation is used to convert Etanamide (CH_3CONH_2) to methylamine (CH_3NH_2). Give the reagent and reaction condition.

a) $Br_2/AgNO_3$ and heat
b) Br_2/KBr and heat
c) Br_2/KOH and heat
d) $Br_2/ethanol$ and reflux

23. Which of the following is NOT a property of d-block elements?

a) They have variable oxidation states
b) They form complexes
c) They have low melting points
d) They form coloured compounds

24. Indicate the type of organic reaction that ethene ($CH_2=CH_2$) and hydrogen chloride could undergo

a) Condensation reaction
b) Nucleophilic addition
c) Electrophilic addition
d) Nucleophilic substitution

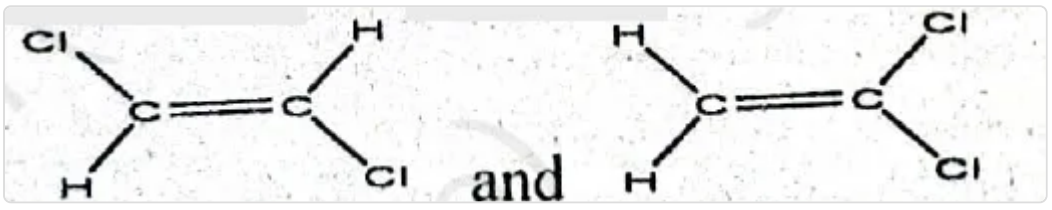
25. Bronsted-Lowry defined a base as

a) an electron pair donor
b) an electron pair acceptor
c) a proton donor
d) a proton acceptor

26. The weakest bond among the following is

a) $F-F$
b) $Cl-Cl$
c) $Br-Br$
d) $Cl-I$

27. The pair of isomers



The diagram shows two molecules of formula $C_2H_2Cl_2$: in the first, one Cl is on each carbon of the $C=C$ bond (on opposite sides, 1,2-dichloroethene); in the second, both Cl atoms are on the same carbon (1,1-dichloroethene, $CH_2=CCl_2$).

are

a) Chain isomers
b) Geometric isomers
c) Position isomers
d) Functional group isomers

28. The first five ionization energies of element X and Y are

X: $500 \;\; 4600 \;\; 6900 \;\; 13400 \;\; 17600$

Y: $790 \;\; 1600 \;\; 3200 \;\; 4400 \;\; 16100.$

A compound formed between X and Y is

Options as printed: A) XY B) X_2Y_3 C) X_3Y_2 D) X_4Y_3

29. In structure determination, the method that identifies the bond type and functional group in the molecule is

a) X-ray diffraction
b) Ultra-violet spectroscopy
c) Infra-red spectroscopy
d) Nuclear magnetic resonance

30. The value of the equilibrium constant for the reaction: $H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$ can be altered by

a) Adding a suitable catalyst
b) Increasing the temperature
c) Increasing the total pressure
d) Adding iodine to the system

31. Which of the compounds below is the most reactive with the nitrating mixture (Conc HNO_3/Conc H_2SO_4)?

a) Benzene
b) Methylbenzene
c) Nitrobenzene
d) Benzoic acid

32. The data below is for the kinetic study of the reaction $A + B \rightarrow C + D$

Conc. A / mol dm^{-3}	**0.40**	**0.28**	**0.20**	**0.14**
Time / sec x 10^2	8	12	16	20

From the data, the order of the reaction is

a) 0
b) 2
c) 3
d) 1

33. The condensed formula for 2-chloro-2-methylpropanal is

a) $\mathrm{CH_3CH(Cl)CHO}$
b) $\mathrm{(CH_3)_2\,C(Cl)\,CHO}$
c) $\mathrm{(CH_3)_2\,C(Cl)\,CH_2\,CHO}$
d) $\mathrm{CH_3CH_2C(CH_3)\,Cl\,CHO}$

34. Which of the following is the correct definition of atomic radius?

a) Half the distance between two identical atoms joined by a single covalent bond
b) Half the distance between the nuclei of two identical atoms joined by a single covalent bond
c) Half the distance between two adjacent atoms of neighbouring molecules which are not chemically bonded
d) Half the distance between two adjacent atoms of neighbouring molecules that are covalently bonded.

35. Which of the chlorides below will react with water to give its corresponding dioxides?

a) NaCl
b) $\mathrm{SiCl_4}$
c) $\mathrm{PCl_3}$
d) $\mathrm{S_2Cl_2}$

36. Which properties of the s-block elements increase down the groups?

1. Electronegativity
2. First ionization energy
3. Metallic character

a) 1, 2 and 3 are all correct
b) 1 and 2 only are correct
c) 2 and 3 only are correct
d) 3 only is correct

37. Hydrogen fluoride is a weaker acid than hydrogen iodide. The reason for this observation is(are),

1. H-F bond is longest and weakest in the halide series
2. H-F bond is shortest and strongest in the halide series
3. It is difficult for HF to release a proton in aqueous solutions

a) 1, 2 and 3 are all correct
b) 1 and 2 only are correct
c) 2 and 3 only are correct
d) 3 only is correct

38. The reaction

$$\mathrm{CH_3CH_2CH_2Cl + KOH \rightarrow CH_3CH_2CH_2OH + KCl}$$

1. Is an example of a bimolecular nucleophilic substitution reaction.
2. Forms an unstable intermediate species
3. Proceeds by the S_N1 mechanism

Options as printed: A) 1, 2 and 3 are all correct B) 1 and 2 only are correct C) 2 and 3 only are correct D) 3 only is correct

39. Which of the following gives a colour change on warming?

1. CH_3CH_2OH and acidified potassium dichromate
2. CH_3CHO and Fehling's solution
3. $(CH_3)_3COH$ and potassium dichromate

a) 1, 2 and 3 are all correct
b) 1 and 2 only are correct
c) 2 and 3 only are correct
d) 3 only is correct

40. A student wanted to measure accurately $10\ cm^3$ of dilute hydrochloric acid. Which of the glassware will you advise the student to use?

1. Volumetric flask
2. Pipette
3. Burette

a) 1, 2 and 3 are all correct
b) 1 and 2 only are correct
c) 2 and 3 only are correct
d) 3 only is correct

41. Some properties of carbon not common to the other elements of group 4 are

1. Covalency of four
2. Catenation
3. Formation of stable multiple bonds

a) 1, 2 and 3 are all correct
b) 1 and 2 only are correct
c) 2 and 3 only are correct
d) 3 only is correct

42. The relative molecular mass of a substance is the

1. Molar mass of the substance without units
2. Mass of a molecule compared to $\frac{1}{12}$th the mass of an atom of the carbon-12 isotope
3. Mass of its molecules compared to $\frac{1}{12}$th the mass of an atom of the carbon-12 isotope.

a) 1, 2 and 3 are all correct
b) 1 and 2 only are correct
c) 2 and 3 only are correct
d) 3 only is correct

43. Identify the element(s) which form(s) hydrides in which hydrogen bonding occur.

1. Oxygen
2. Fluorine
3. Nitrogen

a) 1, 2 and 3 are all correct
b) 1 and 2 only are correct

c) 2 and 3 only are correct
d) 3 only is correct

44. During the industrial preparation of sulphuric acid, it is usually advisable to?

1. Absorb $SO_{3(g)}$ in oleum rather than water
2. Use low temperatures and a catalyst to convert SO_2 to SO_3
3. Use very high pressure to convert SO_2 to SO_3

a) 1, 2 and 3 are all correct
b) 1 and 2 only are correct
c) 2 and 3 only are correct
d) 3 only is correct

45. The following is true about sulphur and its compounds.

1. SO_2 is produced in a stage in the contact process
2. Sulphuric acid is used to manufacture sulphuric acid
3. Sulphur has an oxidation state of 0 (zero) in S_8

Options as printed: A) 1, 2 and 3 are all correct B) 1 and 2 only are correct C) 2 and 3 only are correct D) 3 only is correct

First statement: Primary amines have lower boiling points than their corresponding alcohols of similar molecular weight

Second statement: The intermolecular hydrogen bonds in amines are weaker than in alcohols

a) Both statements are true and the second statement is a CORRECT explanation of the first statement.
b) Both statements are true and the second statement is NOT a CORRECT explanation of the first statement.
c) The first statement is true, but the second statement is false.
d) The first statement is false, but the second statement is true.

First statement: Fluorine is the strongest oxidizing agent in the group of the halogens.

Second statement: Down group VII, the Van der Waal's forces of attraction increase in strength as number of electrons increases.

a) Both statements are true and the second statement is a CORRECT explanation of the first statement.
b) Both statements are true and the second statement is NOT a CORRECT explanation of the first statement.
c) The first statement is true, but the second statement is false.
d) The first statement is false, but the second statement is true.

First statement: Ethanal and propanone differ in their reaction with iodine in sodium hydroxide

Second statement: Ethanal undergoes nucleophilic addition reaction faster than propanone

a) Both statements are true and the second statement is a CORRECT explanation of the first statement.
b) Both statements are true and the second statement is NOT a CORRECT explanation of the first statement.
c) The first statement is true, but the second statement is false.
d) The first statement is false, but the second statement is true.

First statement: Hydrogen gas and hydrogen chloride gas both form hydrogen ion (H^+) when dissolved in water.

Second statement: Hydrogen gas is non-polar while hydrogen chloride gas is polar.

a) Both statements are true and the second statement is a CORRECT explanation of the first statement.
b) Both statements are true and the second statement is NOT a CORRECT explanation of the first statement.
c) The first statement is true, but the second statement is false.
d) The first statement is false, but the second statement is true.

First statement: 2-hydroxypropanoic acid forms two isomers which are mirror images of each other.

Second statement: Organic compounds with an asymmetric carbon atom have the ability to rotate plane polarized light.

a) Both statements are true and the second statement is a CORRECT explanation of the first statement.
b) Both statements are true and the second statement is NOT a CORRECT explanation of the first statement.
c) The first statement is true, but the second statement is false.
d) The first statement is false, but the second statement is true.

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